Using Simple Technical Analysis Indicators for Asset Allocation Decisions

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Abstract

This study analyzes the potential effectiveness of using simple technical analysis indicators to determine the overall riskiness of portfolio asset allocation. Using the 200-day simple moving average of the S&P500 as our technical indicator in two separate strategies, we employ a "risk-on" asset allocation strategy (more portfolio allocation in stocks as a percentage) when the S&P500 is above the indicator line and a "risk-off" strategy (less stock/more bond allocation) when below. We compare the "buy and hold" returns of various portfolio allocations in the U.S. stock and bond markets from 1962-2017. In the initial analysis, we find that following this rule with the 200-day simple moving average provides excess annual returns of up to 0.59%, with all analyzed allocation combinations posting both excess returns and a reduction in overall risk for all analyzed allocation combinations, with all 200-day strategies posting an increase in Sharpe ratios when compared to their baseline "buy-and-hold" strategy.

Keywords: technical analysis; trading strategies; simple moving average

1. Introduction

The effectiveness of technical analysis has been widely analyzed over the years in various asset classes and markets, beginning with Fama and Blume (1966). However, the results of subsequent studies over the next 50-plus years have varied considerably and the effectiveness of using technical analysis still remains an unclear conclusion. Brock et al. (1992) analyzed simple moving averages of the Dow Jones Index from 1897 to 1986 and documented, "strong support for the technical analysis strategies." Hudson et al (1996) found similar results in their UK data set, though they conceded that transaction costs were likely to diminish any excess returns above the buy and hold strategy. Jenson and Benington (1970) also found that technical analysis is not profitable after factoring in trading costs. Allen and Karjalainen (1999) later analyzed technical trading rules against a buy-and-hold strategy using S&P500 stock price data from 1928-1995, and found that technical trading rules do not work after factoring in transaction costs. Fernandez-Rodriguez et al. (2000) utilized technical trading rules based on an artificial networks and found that the technical strategies outperform the buy-and-hold strategies in bear and stable markets, but loses in bull markets. Gorgulho et al. (2011) analyzed more recent data (2003-2009) and found that the technical indicators outperformed versus "buy and hold" and other "random strategies".

Technical analysis trading strategies have yielded positive results in other markets outside of the US stock market. Gunasekarage and Power (2001) found in South Asian markets that moving average trading rules are profitable. Hung and Zhaojun (2013) found similarly positive results when applying a 0.5% transaction cost to their Vietnamese data from 2000-2011.

One crucial aspect of previous technical trading rules literature that we believe hasn't been fully addressed is that most of the previous studies have analyzed technical analysis trading rules as an "all-or-nothing" strategy. As such, an investor either is fully investing in the analyzed assets or not investing at all. It's unclear whether or not these analyses include any return whatsoever when the assets are sold. In reality, when a stock is traded, it can be reallocated to another return yielding asset, such as a bond. Therefore, in this analysis, we propose an alternative to the "all-or-nothing" strategy, and evaluate the effectiveness of changing portfolio asset allocations, both significant and subtle, based on basic technical analysis cues.

Zhu and Zhou (2009) provided a theoretical framework, using technical analysis in order to make better asset allocation decisions. However, it is still somewhat unclear how this framework could be applied by practitioners in the marketplace. This paper seeks to close that gap and provide an empirical analysis on a clear-cut asset allocation strategy based on basic technical analysis. We hope that these insights provide assistance to individuals contemplating the use of technical analysis to influence their asset allocation decisions.

Additionally, we aim to include more recent data set in our analysis, using S&P 500 data from 1962 to 2017. This provides an opportunity to evaluate any changes in the efficacy of the strategies over time. We believe this is important to analyze as recent studies have found evidence that many other "market anomalies" such as momentum investing (Jegadeesh & Titman, 1993), end of the week (French, 1980; Gibbons and Hess, 1981), month and year effects (Keim, 1983; Ariel, 1987), and holiday effects (Ariel, 1990) have either diminished or disappeared in recent years (Dolvin and Foltice, 2017, Robins and Smith, 2016; Robins and Smith, 2017; Robins and Smith, 2019). In this sample, we find that following this rule with the 200-day simple moving average provides excess annual returns of up to 0.59%, with all analyzed allocation combinations posting excess returns. Furthermore, this strategy also shows a reduction in overall risk for all analyzed allocation combinations, with all 200-day strategies posting an increase in Sharpe ratios when compared to the "buy-and-hold" strategy.

2. Methodology

In this analysis, we use daily S&P500 data from January 1, 1962 to December 31, 2017. The 200-day simple moving average line is the average of the previous 200 days closing amounts. We implement a "risk on" portfolio when the 200-day moving average line is below the closing index. The portfolio is switched over to a less risky "risk off" portfolio when the index breaks below the moving average and remains in this portfolio until the index breaks above moving average line.

In this study, we evaluate and compare the index at the end of each day. We analyze a six-by-six (36 in total) matrix of various portfolio asset allocations (100% stocks/0% bonds; 80% stocks/20% bonds; 60% stocks/40% bonds; 40% stocks/60% bonds; 20% stocks/80% bonds; 0% stocks/100% bonds) in order to test whether this trading strategy has been historically effective in either (or both) increasing returns and/or reducing portfolio risk.

In the initial analysis, we change the portfolio composition at the end of the following day when the index breaks above/below the technical indicator line. For example, if the index crosses below the indicator line at the end of the day on a Monday, we wait until the end of the day on Tuesday to switch the portfolio from a "risk on" portfolio to a "risk off" portfolio. If the index crosses back above the indicator line at the end of the day on Tuesday, we switch the portfolio composition "on" at the end of the following day (Wednesday).

When the opposite occurs, i.e. the index breaks above the indicator line at the end of the day on a Monday, we wait until the end of the day Tuesday to change the portfolio composition to a "risk on" portfolio. If the index breaks below the indicator line at the end of the day on that Tuesday, we switch the portfolio over at the end of the day on Wednesday.

For the bond portfolios, we use monthly 10-year T-Bond yields and returns provided by Kenneth French's Data Library (Kenneth R. French – Data Library, 2018). In this analysis, we cut the monthly yields and returns to daily returns. Similarly, we use monthly historical S&P050 dividend yields and divide these into daily returns for our analysis. These were the only data that we could retrieve back to 1962. We strongly believe that, qualitatively, this does not play a significant role in our results and overall findings.

3. Results – Main Analysis

Table 1 shows the "buy and hold" results of the six different portfolio allocation mixes analyzed throughout this paper. This serves as benchmark to compare our trading strategies by. For the Sharpe ratios in this analysis, we use a risk free rate of 5.1%, which is the average 1 year US T-Bill yield over the entire sample period.

Table 1. Benchmark "Buy and Hold" Portfolio Results – Overall Sample Period (1962-2017)

Stock/Bond %'s	***100%/0%***	***80%/20%***	***60%/40%***	***40%/60%***	***20%/80***	***0%/100%***
Avg. Return	10.03%	9.44%	8.76%	8.09%	7.42%	6.75%
St. Dev.	0.1647	0.1354	0.1159	0.0994	0.1015	0.1173
*Sharpe Ratio	0.3290	0.3506	0.3627	0.3406	0.2675	0.1742

Note: This table displays the overall sample results of the six various "buy and hold" portfolio asset allocation mixes examined in this analysis.
*Risk free rate = 5.1%

In our main analysis, we implement a six by five matrix of various "risk on" and "risk off" measures.[1] Based on the results shown in Table 2, it appears that implementing this strategy generates not only excess returns, but would concurrently reduce volatility. Sharpe ratios are the highest for all "risk off" portfolios when they are cut down 0% stocks when the indicator line is above the index. The highest Sharpe ratio posted is the most extreme strategy, which is a 100%

[1] We do not include the 0%/100% "risk on" and 0%/100% "risk off" strategy in the analyses.

stock “risk on” portfolio and 0% stock “risk off” portfolio. Here, we find that over the entire sample period this strategy would require an average of 6.23 trades a year, ranging from 0 up to 21 times.

Table 2. Trading Strategy 1. Main Analysis 200-Day SMA - “Risk-On” Portfolio / “Risk-Off” Portfolio Overall Sample Period (1962-2017)

Avg. Trades/year = 6.23 Min = 0 Max = 21

	100%/0%	**80%/20%**	**60%/40%**	**40%/60%**	**20%/80%**	**0%/100%**
100%/0%						
Avg. Return	10.03%	10.15%	10.27%	10.39%	10.51%	10.62%
St. Dev.	0.1647	0.1520	0.1416	0.1342	0.1308	0.1319
Sharpe Ratio	0.3313	0.3668	0.4023	0.4332	0.4533	0.4579
80%/20%						
Avg. Return		8.94%	9.01%	9.12%	9.24%	9.34%
St. Dev.		0.1337	0.1217	0.1130	0.1087	0.1098
Sharpe Ratio		0.3265	0.3640	0.4025	0.4286	0.4342
60%/40%						
Avg. Return			7.85%	7.89%	8.00%	8.10%
St. Dev.			0.1057	0.0958	0.0907	0.0919
Sharpe Ratio			0.3097	0.3455	0.3772	0.3837
40%/60%						
Avg. Return				6.76%	6.79%	6.90%
St. Dev.				0.0838	0.0785	0.0800
Sharpe Ratio				0.2605	0.2818	0.2899
20%/80%						
Avg. Return					5.67%	5.72%
St. Dev.					0.0736	0.0758
Sharpe Ratio					0.1483	0.1505

Note: This table outlines the returns, volatility, and Sharpe ratios of all “risk on” (left column) and “risk off” (rows) portfolio stock and bond asset allocation (as a percentage) for the entire analyzed sample, from 1962-2017. We use the simple 200-day moving average as the indicator line to dictate the portfolio allocation at the end of the following trading day.

In the next section, we split the time series into two equal sub-periods, 1962-1989 and 1990-2017, in order to investigate if the returns have diminished over time. In Table 3, we can see that the “risk on/risk off” strategy is quite effective, both from an excess return standpoint (up to 1.28% annually) and a reduction of volatility. This combination leads to an increase in Sharpe ratios up to 52.41% (for the 100%/0% “risk on” and 0%/100% “risk off” strategy).

Table 3. Main Analysis - 200-Day SMA - "Risk-On" Portfolio / "Risk-Off" Portfolio Sub-Period 1 (1962-1989) – Stock/Bond Portfolio Mix

	100%/0%	80%/20%	60%/40%	40%/60%	20%/80%	0%/100%
100%/0%						
Avg. Return	9.06%	9.32%	9.58%	9.84%	10.09%	10.34%
St. Dev.	0.1579	0.1470	0.1387	0.1334	0.1314	0.1329
Sharpe Ratio	0.2866	0.3257	0.3640	0.3978	0.4230	0.4368
80%/20%						
Avg. Return		8.15%	8.36%	8.61%	8.86%	9.10%
St. Dev.		0.1323	0.1218	0.1148	0.1186	0.1123
Sharpe Ratio		0.2736	0.3146	0.3554	0.3876	0.4063
60%/40%						
Avg. Return			7.25%	7.43%	7.67%	7.90%
St. Dev.			0.1103	0.1017	0.0972	0.0972
Sharpe Ratio			0.2461	0.2848	0.3224	0.3463
40%/60%						
Avg. Return				6.34%	6.51%	6.74%
St. Dev.				0.0945	0.0894	0.0887
Sharpe Ratio				0.1916	0.2213	0.2483
20%/80%						
Avg. Return					5.44%	5.61%
St. Dev.					0.0882	0.0877
Sharpe Ratio					0.1027	0.1230

Note: This table outlines the returns, volatility, and Sharpe ratios of all "risk on" (left column) and "risk off" (rows) portfolio stock and bond asset allocation (as a percentage) for sub-period #1, from 1962-1989. We use the simple 200-day moving average as the indicator line to dictate the portfolio allocation at the end of the following trading day.

In Table 4, when we analyze sub-period #2 (1990-2017), we find that the magnitude of the results have diminished. Here, the returns for this strategy are actually marginally less than the "buy-and-hold" strategies across all strategies, for example, -15 basis points for the (for the 60%/40% "risk on" and 0%/100% "risk off" strategy). However, the trading strategy maintains a widespread reduction of portfolio volatility over this time. The Sharpe ratio for the extreme 100%/0% "risk on" and 0%/100% "risk off" strategy posts a 28.35% increase during this sub-period.

Table 4. Main Analysis - 200-Day SMA - "Risk-On" Portfolio / "Risk-Off" Portfolio Sub-Period 2 (1990-2017) – Stock/Bond Portfolio Mix

	100%/0%	80%/20%	60%/40%	40%/60%	20%/80%	0%/100%
100%/0%						
Avg. Return	11.01%	10.99%	10.97%	10.95%	10.92%	10.89%
St. Dev.	0.1735	0.1591	0.1466	0.1373	0.1325	0.1332
Sharpe Ratio	0.4565	0.4964	0.5372	0.5722	0.5913	0.5859
80%/20%						
Avg. Return		9.73%	9.65%	9.63%	9.62%	9.59%
St. Dev.		0.1370	0.1235	0.1129	0.1076	0.1092
Sharpe Ratio		0.4849	0.5314	0.5798	0.6063	0.5955
60%/40%						
Avg. Return			8.46%	8.35%	8.33%	8.31%
St. Dev.			0.1025	0.0912	0.0854	0.0881
Sharpe Ratio			0.5233	0.5765	0.6140	0.5930
40%/60%						
Avg. Return				7.18%	7.07%	7.06%
St. Dev.				0.0731	0.0676	0.0718
Sharpe Ratio				0.5594	0.5895	0.5528
20%/80%						
Avg. Return					5.90%	5.83%
St. Dev.					0.0570	0.0633
Sharpe Ratio					0.4930	0.4322

Note: This table outlines the returns, volatility, and Sharpe ratios of all "risk on" (left column) and "risk off" (rows) portfolio stock and bond asset allocation (as a percentage) for sub-period #2, from 1990-2017. We use the simple 200-day moving average as the indicator line to dictate the portfolio allocation at the end of the following trading day.

4. Results – Analysis #2

In the second analysis, we wait for the index to stay above/below the indicator line for two consecutive days before switching the portfolio composition on the third day. For example, if the index decreases and crosses below the indicator line at the end of the day on a Monday, we wait until the end of the day on Tuesday to confirm that the index remains below the indicator line before switching the portfolio from a "risk on" portfolio to a "risk off" portfolio at the end of the day on Wednesday. If the index line breaks back above the indicator line on that Tuesday, the portfolio composition doesn't change.

This "two day rule" not only reduces excessive trading (and the associated trading costs), it also provides a more realistic analysis for portfolio managers, who likely, may not have the capability to switch their clients' portfolio asset allocation back and forth throughout the trading week.

In Table 5, we find that this trading rule reduces average annual trades from 6.23 down to 3.39, a reduction of 45.59%. Furthermore, we find that the returns are comparatively enhanced, yielding excess returns up to 0.79% annually. Volatility is, again, reduced throughout all analyzed combinations with increases in the Sharpe ratio up to 46.06%.

Table 5. 200-Day SMA - "Risk-On" Portfolio / "Risk-Off" Portfolio Overall Sample Period (1962-2017)

Two-Day Rule – Avg. Trades/year = 3.39 Min = 0 Max = 9

	100%/0%	**80%/20%**	**60%/40%**	**40%/60%**	**20%/80%**	**0%/100%**
100%/0%						
Avg. Return	10.03%	10.20%	10.36%	10.52%	10.67%	10.82%
St. Dev.	0.1647	0.1516	0.1406	0.1327	0.1286	0.1291
Sharpe Ratio	0.3313	0.3707	0.4110	0.4476	0.4739	0.4839
80%/20%						
Avg. Return		8.94%	9.05%	9.21%	9.37%	9.52%
St. Dev.		0.1337	0.1212	0.1120	0.1071	0.1076
Sharpe Ratio		0.3265	0.3689	0.4137	0.4471	0.4591
60%/40%						
Avg. Return			7.85%	7.93%	8.09%	8.24%
St. Dev.			0.1057	0.0954	0.0898	0.0904
Sharpe Ratio			0.3097	0.3516	0.3910	0.4048
40%/60%						
Avg. Return				6.76%	6.84%	6.99%
St. Dev.				0.0838	0.0782	0.0793
Sharpe Ratio				0.2605	0.2888	0.3040
20%/80%						
Avg. Return					5.67%	5.77%
St. Dev.					0.0736	0.0757
Sharpe Ratio					0.1483	0.1569

Note: This table outlines the returns, volatility, and Sharpe ratios of all "risk on" (left column) and "risk off" (rows) portfolio stock and bond asset allocation (as a percentage) for the entire analyzed sample, from 1962-2017. We use the simple 200-day moving average as the indicator line to dictate the portfolio allocation. Before switching the portfolio allocation in this analysis on the third day, the index must stay above/below the indicator line for two consecutive days.

In Table 6, we analyze sub-period #2, from 1990-2017, for the "Two-Day" trading rule. Here, we find in general, this strategy increases returns and simultaneously decreases volatility. With the exception of the 20/80 risk on and 0/100 risk off portfolio, we see increases in the Sharpe ratio up to 41.03%.

Table 6. 200-Day SMA – Two-Day Rule - "Risk-On" Portfolio / "Risk-Off" Portfolio: Sub-Period #2 (1990-2017)

	100%/0%	80%/20%	60%/40%	40%/60%	20%/80%	0%/100%
100%/0%						
Avg. Return	11.01%	11.06%	11.11%	11.16%	11.21%	11.25%
St. Dev.	0.1735	0.1582	0.1447	0.1340	0.1276	0.1267
Sharpe Ratio	0.4565	0.5037	0.5545	0.6025	0.6313	0.6438
80%/20%						
Avg. Return		9.73%	9.73%	9.79%	9.84%	9.89%
St. Dev.		0.1370	0.1226	0.1109	0.1042	0.1042
Sharpe Ratio		0.4849	0.5413	0.6038	0.6478	0.6520
60%/40%						
Avg. Return			8.46%	8.42%	8.49%	8.54%
St. Dev.			0.1025	0.0904	0.0835	0.0849
Sharpe Ratio			0.5233	0.5902	0.6468	0.6424
40%/60%						
Avg. Return				7.18%	7.15%	7.22%
St. Dev.				0.0731	0.0670	0.0704
Sharpe Ratio				0.5594	0.6066	0.5864
20%/80%						
Avg. Return					5.90%	5.91%
St. Dev.					0.0570	0.0633
Sharpe Ratio					0.4930	0.4458

Note: This table outlines the returns, volatility, and Sharpe ratios of all "risk on" (left column) and "risk off" (rows) portfolio stock and bond asset allocation (as a percentage) for sub-period #2, from 1990-2017. We use the simple 200-day moving average as the indicator line to dictate the portfolio allocation. Before switching the portfolio allocation in this analysis on the third day, the index must stay above/below the indicator line for two consecutive days.

5. Discussion/Conclusion

This paper analyzes the potential effectiveness of using simple technical analysis indicators to determine the riskiness levels of portfolio asset allocation. Using the 200-day simple moving average of the S&P500 as our technical indicator in two separate strategies, we employ a "risk-on" asset allocation strategy (more portfolio allocation in stocks as a percentage) when the S&P500 is above the indicator line and a "risk-off" strategy (less stock/more bond allocation) when below. We compare the "buy and hold" returns of various portfolio allocations in the U.S. stock and bond markets from 1962-2017. In our initial analysis, we find that following this rule with the 200-day simple moving average provides excess annual returns of up to 0.59%, with all analyzed allocation combinations posting excess returns. Furthermore, this strategy also shows a reduction in overall risk for all analyzed allocation combinations, with all 200-day strategies posting an increase in Sharpe ratios when compared to their baseline "buy-and-hold" strategy. However, it appears that some of the effectiveness of this strategy has diminished over time. The excess returns for our more recent sub-period (1990-2017) show a reduction of annual returns up to 15 basis points. Nevertheless, the benefits of the reduction of risk using this trading strategy appear to outweigh the small reduction in returns, as depicted by the consistently higher Sharpe ratios across all portfolios.

In a second analysis seeking to reduce the number of trades, we find even great excess returns as well as reduced volatility, in the overall sample as well as both sub-periods. We did not include transaction costs in this analysis. Previous literature has shown that the effectiveness is diminished by transaction cost (Jensen & Benington, 1970; Hudson, 1996; Allen & Karjalainen, 1999). While that may have been true in the past, we believe that switching portfolio asset allocation through the use of ETF's today can be a no/low cost solution to implement this strategy.

We believe that practitioners and retail investors could benefit from implementing this strategy or a similar variation of it. At the very least, using a simple technical indicator could reduce emotion in turbulent markets, providing clear stock market entry and exit points to dictate when to change a portfolio.

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